

ELEC 3117

Electrical Engineering Design

For this activity, in the lab in week 3, you will need to design and implement a power supply for your project. This activity aims to introduce you to the basic concepts of voltage regulation, current limiting, and power conversion as well as the fundamentals of PCB design and fabrication. It will provide you with a hands-on experience on how to create circuit schematics and translate them into PCB layouts using design software. It will also offer you the opportunity to learn practical skills such as soldering, component placement, routing traces and grounding techniques on a PCB in addition to the testing procedures required to validate the functionality and performance of your designed power supply.

In your design, you will need to use one or more of the components listed in the Table below. To design your power supply, you will need to:

- Use an input voltage of 12V.
- Determine the power requirements (voltage, current) for your project.
- Design a suitable power supply to generate the required voltage with an appropriate current output.
- Layout a 48x48mm PCB of your designed circuit using the template provided on Moodle. All connections to and off the PCB are to be made using standard 0.254mm pitch header pins.
- Include your name and group number in the silkscreen of your PCB.
- Test that your board uploads correctly at <https://ilcpcb.com/> (You should not order it here, just ensure all layers and the drill holes are detected)

IC	Inductor values
AZ34063UMTR-G1	SRR1260-221K
MC33063ADR	RLB0914-820KL
MC34063ECN	SRN8040TA-820M
	RCH8011NP-221L

At the end of the lab, you will need to have your design and PCB layout reviewed by the demos. You can then fix any identified mistakes based on the feedback received.

Each group will need to submit a 3-page design report on Moodle that includes all relevant calculations, schematic and corrected Gerber files of the PCB layout for printing. The School Workshop will order the PCBs. When received, you will need to collect your PCB and the required components from the workshop (cost ~\$10). **You should not order the PCB or components yourself!** You will then need to assemble and demonstrate your working power supply PCB in the week 7 progress review.

You can use your power supply PCB in your final project demonstration in Week 10 but it will not count toward the bonus marks for a PCB of your final product. If you choose to make a PCB of your final project for the bonus marks, you can also add your designed PCB to the final board instead of using your power supply PCB